Chapter 7. Hypothesis Testing

Logic of tests, likelihood ratio tests, large-sample approximations

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7.1 The logic of tests

A test of $H_0 : \theta \in \Omega_0$ against $H_1 : \theta \in \Omega_1$ ($\Omega_0 \cap \Omega_1 = \emptyset$) is a rule that rejects H_0 when the data fall in a **critical region** C. Equivalently, a test is a function $\phi(\mathbf{x}) \in [0,1]$ giving the probability of rejecting; $\phi = I_C$ for a non-randomized test.

- **Type I error:** rejecting a true H_0. **Type II error:** accepting a false H_0.
- The **power function** is $\pi(\theta) = E_\theta \phi(\mathbf{X}) = P_\theta(\text{reject})$.
- The **size** is $\sup_{\theta \in \Omega_0} \pi(\theta)$. A test has **level** α if its size is $\le \alpha$.

The two errors cannot both be made small for fixed n. The convention is to bound the type I error by α and then make the power on Ω_1 as large as possible, so rejecting H_0 is the strong conclusion.

Randomized tests. With discrete data an exact size α may require randomizing on the boundary: reject when $T > c$, and with probability γ when $T = c$, where $P_{\theta_0}(T > c) + \gamma P_{\theta_0}(T = c) = \alpha$.

Example 7.1

$X \sim B(10, p)$, $H_0 : p = \frac{1}{2}$ against $p > \frac{1}{2}$, $\alpha = 0.05$. Here $P(X \ge 9) = 11/1024 = 0.0107$ and $P(X = 8) = 45/1024 = 0.0439$. The randomized test rejects when $X \ge 9$, and with probability $\gamma = (0.05 - 0.0107)/0.0439 = 0.893$ when $X = 8$.

p-values. For a test rejecting when $T \ge c$, the p-value is $\sup_{\theta \in \Omega_0} P_\theta(T \ge t_{\text{obs}})$; rejecting when $p \le \alpha$ is a level-α test.

7.2 Likelihood ratio tests

Definition (Likelihood ratio test)

With $\Omega = \Omega_0 \cup \Omega_1$ and MLEs $\hat\theta^{\Omega}$, $\hat\theta^{\Omega_0}$ over Ω and Ω_0, the LRT rejects H_0 for large

$$\Lambda = \frac{\max_{\Omega} L(\theta)}{\max_{\Omega_0} L(\theta)}, \qquad \text{i.e.} \qquad 2\log\Lambda = 2\{l(\hat\theta^{\Omega}) - l(\hat\theta^{\Omega_0})\} \ge c,$$

with c chosen so that the size is α.

Since $\max_\Omega L = \max\{\max_{\Omega_0} L, \max_{\Omega_1} L\}$, a large ratio means the data fit some $\theta \in \Omega_1$ much better than any $\theta \in \Omega_0$. In practice one simplifies the critical region to an equivalent statistic with a known null distribution.

Example 7.2 (Normal mean, σ unknown)

For $N(\mu, \sigma^2)$ and $H_0 : \mu = \mu_0$, $\hat\sigma^2_\Omega = \frac{1}{n}\sum(x_i - \bar x)^2$ and $\hat\sigma^2_0 = \frac{1}{n}\sum(x_i - \mu_0)^2 = \hat\sigma^2_\Omega + (\bar x - \mu_0)^2$. So

$$2\log\Lambda = n\log\frac{\hat\sigma^2_0}{\hat\sigma^2_\Omega} = n\log\Big(1 + \frac{T^2}{n-1}\Big), \qquad T = \frac{\sqrt{n}(\bar x - \mu_0)}{s}.$$

This is increasing in $|T|$, so the LRT is the two-sided t-test: reject when $|T| \ge t_{\alpha/2}(n-1)$.

Example 7.3 (Normal variance)

For $H_0 : \sigma^2 = \sigma_0^2$ with μ unknown, with $u = \sum(x_i - \bar{x})^2/\sigma_0^2$ we get $2\log\Lambda = u - n - n\log(u/n)$. This is convex in u with minimum at $u = n$, so the LRT rejects when $u \le c_1$ or $u \ge c_2$. Under H_0, $u \sim \chi^2(n-1)$; equal tails $c_1 = \chi^2_{1-\alpha/2}(n-1)$, $c_2 = \chi^2_{\alpha/2}(n-1)$ give the usual approximation to the exact LRT.

Other standard LRTs reduce in the same way: two-sample t (equal variances), the F-test for two variances, and binomial or Poisson tests (randomized for exactness).

7.3 Large-sample approximations

Theorem 7.1 (Wilks, simple null)

Under the regularity conditions of Theorem 6.6 with $\Omega \subseteq \mathbb{R}^k$ open, if $H_0 : \theta = \theta_0$ is true,

$$2\{l(\hat{\theta}_n) - l(\theta_0)\} \xrightarrow{d} \chi^2(k).$$

Proof. Taylor-expand around $\hat{\theta}_n$, where $\nabla l(\hat{\theta}_n) = 0$:

$$l(\theta_0) = l(\hat{\theta}_n) + \tfrac{1}{2}(\theta_0 - \hat{\theta}_n)'\nabla^2 l(\theta^*)(\theta_0 - \hat{\theta}_n).$$

By consistency and the LLN, $-\frac{1}{n}\nabla^2 l(\theta^*) \xrightarrow{p} I(\theta_0)$. Hence $2\{l(\hat{\theta}_n) - l(\theta_0)\} = \sqrt{n}(\hat{\theta}_n - \theta_0)'\{I(\theta_0) + o_p(1)\}\sqrt{n}(\hat{\theta}_n - \theta_0)$. With $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} Z \sim N_k(\mathbf{0}, I^{-1})$, the limit is $Z'IZ \sim \chi^2(k)$ (quadratic forms of normal vectors). ∎

Theorem 7.2 (Wilks, composite null)

Partition $\theta = (\xi, \eta)$ with $\xi \in \mathbb{R}^{k_1}$, and test $H_0 : \xi = \xi_0$ (η unrestricted, $k_0 = k - k_1$ free parameters). Under H_0,

$$2\{l(\hat{\theta}_n^{\Omega}) - l(\hat{\theta}_n^{0})\} \xrightarrow{d} \chi^2(k_1),$$

and the **Wald** and **score** (Rao) statistics

$$W_n = n(\hat{\xi}_n - \xi_0)'\big[\{I(\hat{\theta}_n)^{-1}\}_{11}\big]^{-1}(\hat{\xi}_n - \xi_0), \qquad R_n = \tfrac{1}{n}\nabla l(\hat{\theta}_n^0)' I(\hat{\theta}_n^0)^{-1}\nabla l(\hat{\theta}_n^0)$$

have the same $\chi^2(k_1)$ limit, and all three differ by $o_p(1)$.

Sketch. Write $I = \begin{pmatrix} I_{11} & I_{12} \\ I_{21} & I_{22} \end{pmatrix}$ and $I_{11\cdot 2} = I_{11} - I_{12}I_{22}^{-1}I_{21}$. Expansions as in Theorem 7.1 give $2\{l(\hat{\theta}^{\Omega}) - l(\theta_0)\} \approx \mathbf{U}'I^{-1}\mathbf{U}$ and $2\{l(\hat{\theta}^{0}) - l(\theta_0)\} \approx \mathbf{U}_2'I_{22}^{-1}\mathbf{U}_2$, where $\mathbf{U} = n^{-1/2}\nabla l(\theta_0) \xrightarrow{d} N(\mathbf{0}, I)$. Their difference is $\mathbf{Z}'I_{11\cdot 2}^{-1}\mathbf{Z}$ with $\mathbf{Z} = \mathbf{U}_1 - I_{12}I_{22}^{-1}\mathbf{U}_2 \xrightarrow{d} N(\mathbf{0}, I_{11\cdot 2})$ (block-inverse formula), so it is $\chi^2(k_1)$. Since $\sqrt{n}(\hat{\xi} - \xi_0) \approx I_{11\cdot 2}^{-1}\mathbf{Z}$ and $\{I^{-1}\}_{11} = I_{11\cdot 2}^{-1}$, W_n has the same leading term. For R_n, $\nabla_\eta l(\hat{\theta}^0) = 0$ and $n^{-1/2}\nabla_\xi l(\hat{\theta}^0) \approx \mathbf{Z}$. ∎

A null of the form $g_1(\theta) = \cdots = g_{k_1}(\theta) = 0$ is handled by reparametrizing so that these functions become coordinates.

Application: multinomial goodness of fit. For Multi$(n, \mathbf{p})$ and $H_0 : \mathbf{p} = \mathbf{p}(\eta)$ with $\eta \in \mathbb{R}^{k_0}$ (and $k-1$ free cell probabilities under Ω), Wilks gives

$$G^2 = 2\sum_j x_j \log\frac{x_j}{n\hat{p}_j^0} \xrightarrow{d} \chi^2(k - 1 - k_0),$$

and the Pearson statistic $X^2 = \sum_j \frac{(x_j - n\hat{p}_j^0)^2}{n\hat{p}_j^0}$ is asymptotically equivalent. (Expand $x \log \frac{x}{e} \approx (x-e) + \frac{(x-e)^2}{2e}$ and note $\sum(x_j - e_j) = 0$.)

Example 7.4 (Independence in an $r \times c$ table)

$H_0 : p_{ij} = p_{i\cdot} p_{\cdot j}$, with MLEs $\hat{p}_{i\cdot} = x_{i\cdot}/n$ and $\hat{p}_{\cdot j} = x_{\cdot j}/n$. The degrees of freedom are $(rc-1)-(r-1)-(c-1) = (r-1)(c-1)$. For the 2×2 table with rows $(30, 20)$ and $(20, 30)$, all expected counts are 25, so $X^2 = 4 \cdot \frac{25}{25} = 4.0 > \chi^2_{0.05}(1) = 3.84$: reject independence at the 5% level.